



Customer Segmentation and Behavioral Analysis Using the Online Retail Dataset

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Data Science Master's
Project

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Related Work and Motivation

- Prior work used **RFM + k-means** for customer segmentation
- Analysis focused mainly on **aggregated RFM features**
- Limited forecasting and product relationship analysis
- Goal: integrate clustering, prediction, forecasting, and association rules into one workflow

Traditional RFM Approach
RFM
K-means
Segmentation Only



Integrated Customer Analytics Workflow
Clustering
Regression
Forecasting
Association Rules
Evaluation Metrics

RECENCY:
Time since
last order

FREQUENCY:
Total # of
transactions

MONETARY:
Total
transaction
value

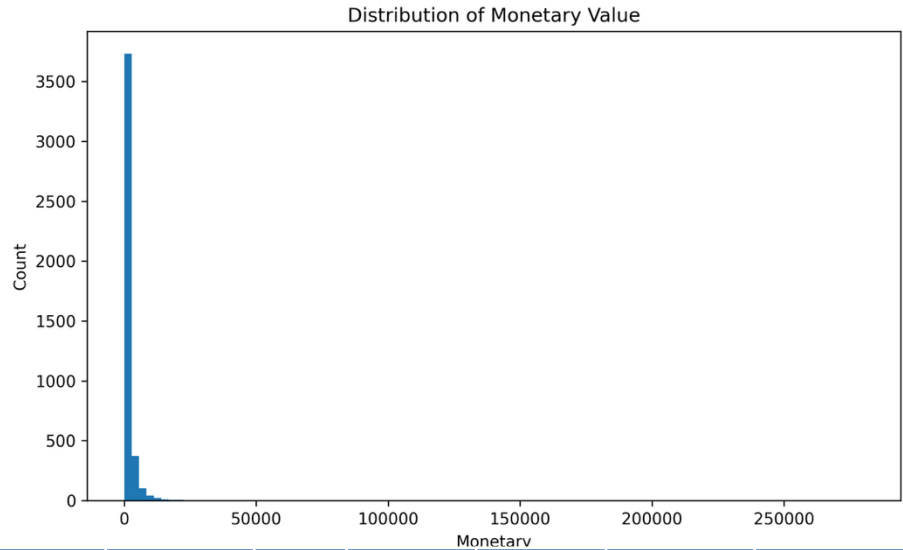
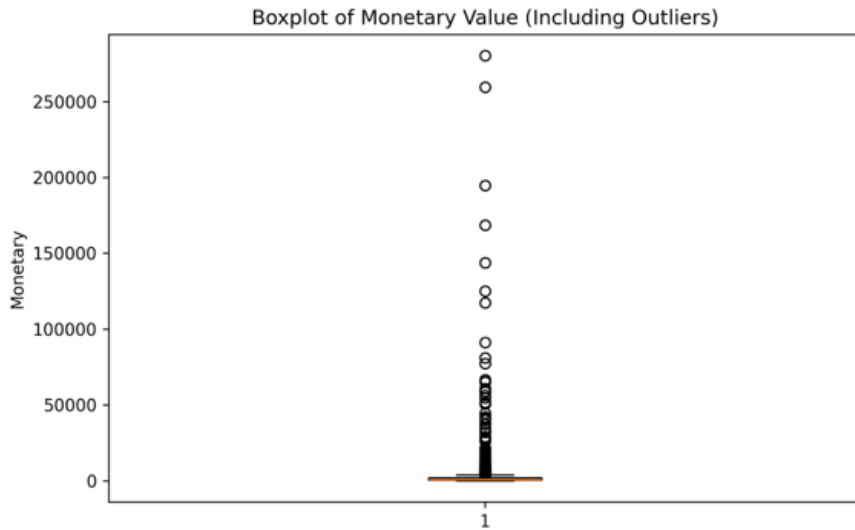
Dataset & Features

- UCI Online Retail Dataset
- 541,909 transaction records
- RFM variables: **Recency, Frequency, Monetary**
- Additional features: **Product diversity and transaction timing**
- Preprocessing removed invalid and missing records

Attribute	DataType	ExampleValue	Description
InvoiceNo	object	536365	Unique transaction ID
StockCode	object	85123A	Product code
Description	object	WHITE HANGING HEART T-LIGHT HOLDER	Product description
Quantity	int64	6	Quantity of product purchased
InvoiceDate	datetime64[ns]	2010-12-01 08:26:00	Date and time of transaction
UnitPrice	float64	2.55	Price per unit
CustomerID	float64	17850.0	Unique customer identifier
Country	object	United Kingdom	Country of customer

Column	MissingCount
Description	1454
CustomerID	135,080

Exploratory Data Analysis & Preprocessing

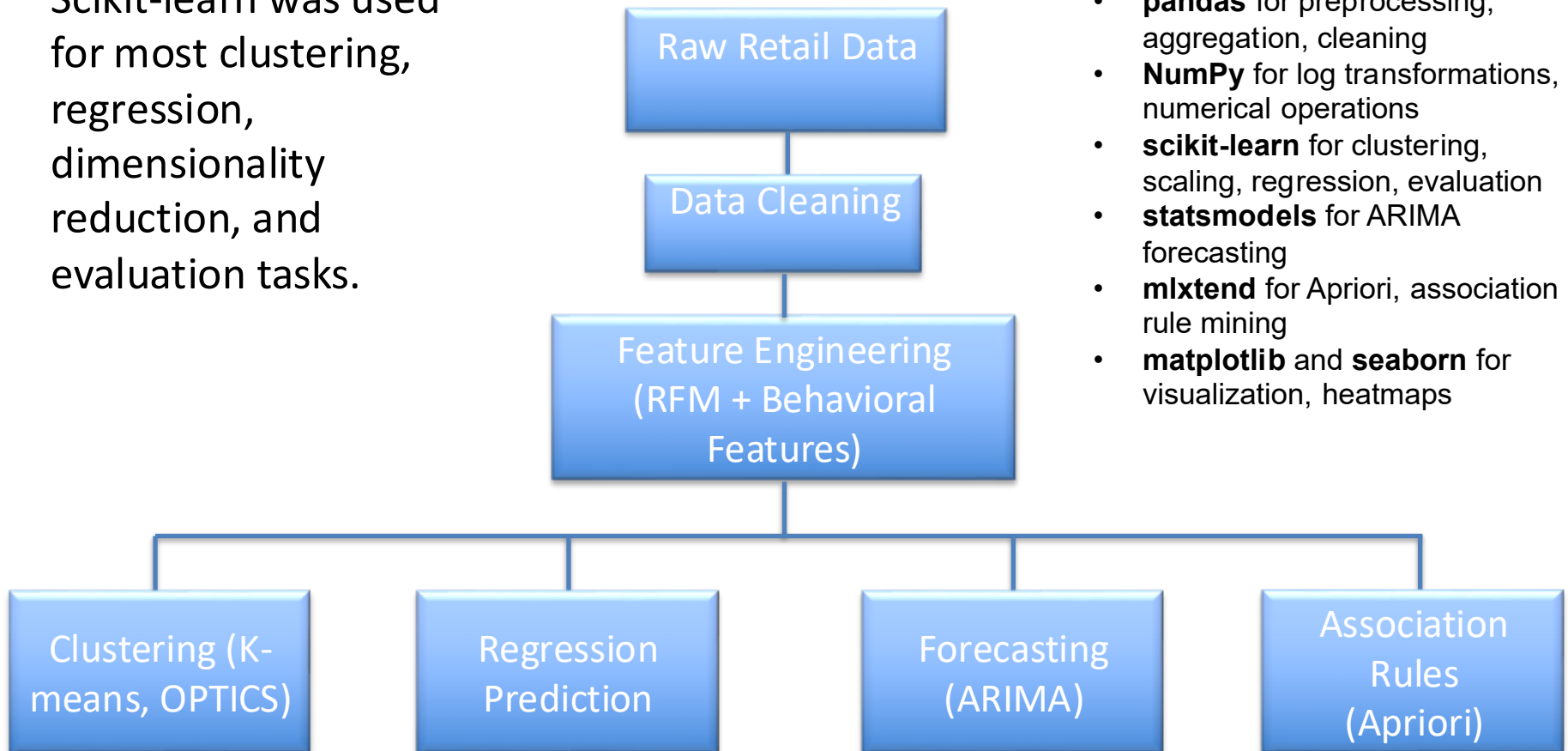


Attributes	count	mean	std	min	25%	50%	75%	max
AvgPurchaseValue	4338.0	419.17	1796.54	3.45	178.62	293.9	430.11	84236.25
ProductDiversity	4338.0	61.5	85.37	1.0	16.0	35.0	77.0	1787.0
AvgDaysBetweenTransactions	4338.0	47.34	63.18	0.0	0.0	28.0	68.0	365.0
Recency	4338.0	92.54	100.01	1.0	18.0	51.0	142.0	374.0
Frequency	4338.0	4.27	7.7	1.0	1.0	2.0	5.0	209.0
Monetary	4338.0	2054.27	8,989.23	3.75	307.41	674.48	1661.74	280206.02

- Monetary distribution showed strong right skew
- Extreme spending outliers compressed most customer patterns
- Customer behavior varied substantially across the dataset

Methodology Pipeline

Scikit-learn was used for most clustering, regression, dimensionality reduction, and evaluation tasks.

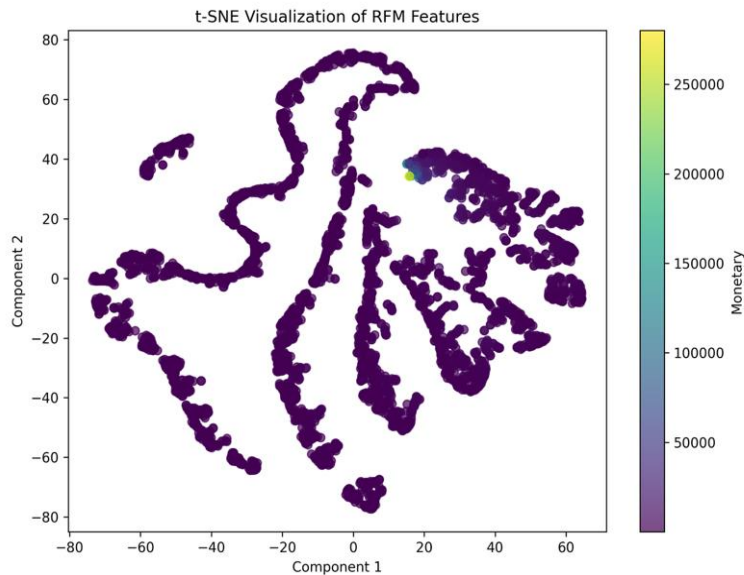


Key Python Libraries and Uses

- **pandas** for preprocessing, aggregation, cleaning
- **NumPy** for log transformations, numerical operations
- **scikit-learn** for clustering, scaling, regression, evaluation
- **statsmodels** for ARIMA forecasting
- **mlxtend** for Apriori, association rule mining
- **matplotlib** and **seaborn** for visualization, heatmaps

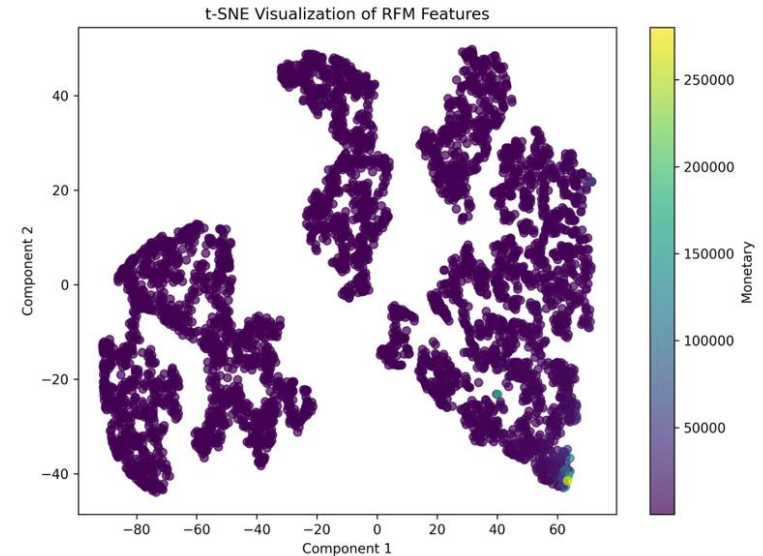
Clustering Visualization and Log Transformation

Before preprocessing t-SNE



Raw customer data showed heavy skew and overlapping structure

After preprocessing and scaling



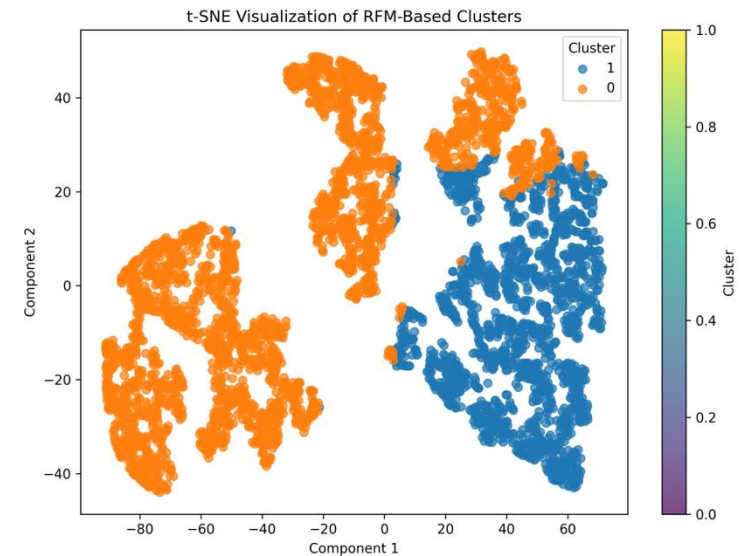
Log transformation reduced the impact of extreme values

- Customer spending behavior was highly skewed
- Log transformation reduced extreme value compression
- Preprocessing improved cluster separability

Clustering Evaluation Results

- Best clustering performance occurred at k=2
- Traditional RFM produced stronger separation than extended features
- OPTICS produced weaker cluster separation

Method	Clusters	Silhouette Score	DaviesBouldin Index
KMeans_RFM	2	0.4329	0.8921
KMeans_RFM	3	0.3365	1.048
KMeans_RFM	4	0.3371	1.0101
KMeans_RFM	5	0.3161	0.9895
KMeans_RFM	6	0.3133	1.0134
KMeans_RFM	7	0.31	0.9726
KMeans_RFM	8	0.3008	0.994

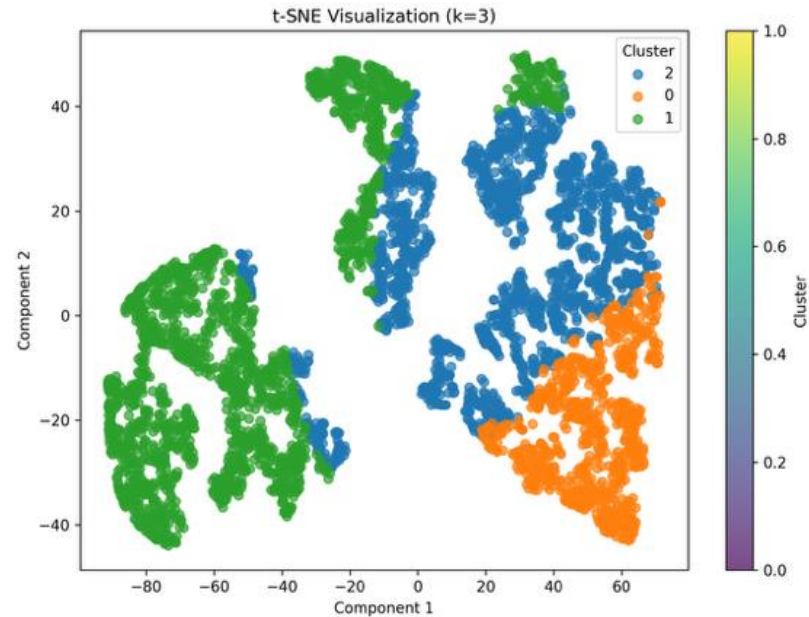


Cluster _RFM	Interpretation	Recency	Frequency	Monetary	Customer Count	Percentage
0	Inactive/Low Value Customers	134.14	1.67	497.74	2671	61.57
1	Active high-value customers	25.88	8.44	4548.26	1667	38.43

Customer Segment Interpretation

k=3 was explored for more detailed behavioral segmentation

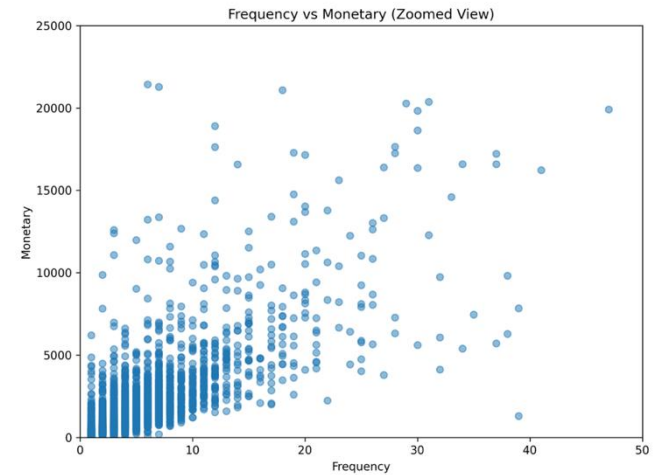
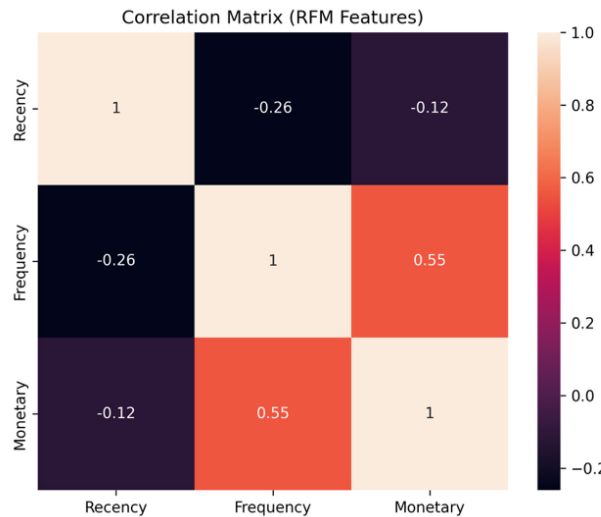
- Cluster 0 contained the highest-spending and most active customers
- Cluster 1 represented inactive customers with low purchase frequency
- Cluster 2 showed moderate purchasing behavior and engagement



Cluster	Interpretation	Recency	Freq	Monetary	Customer Count	Percentage
3						
0	High-value active customers	17.06	13.34	7,905.44	770	17.75
1	Inactive low-spending customers	167.36	1.35	362.54	1872	43.15
2	Moderate-value regular customers	44.21	3.38	1,265.06	1696	39.1

Prediction Modeling

- Frequency showed the strongest relationship with spending
- Extreme outliers reduced regression performance
- Additional behavioral features did not improve prediction accuracy



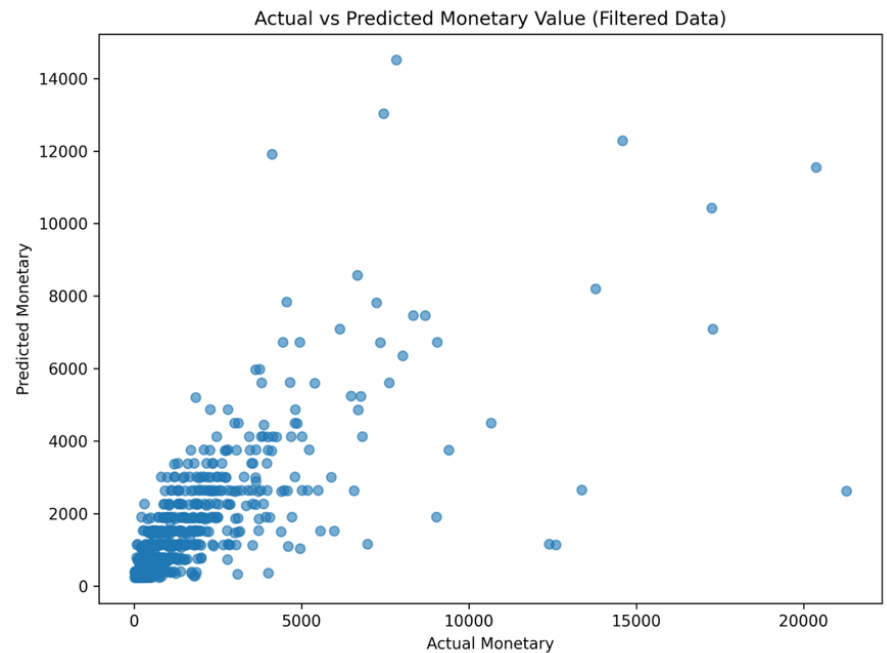
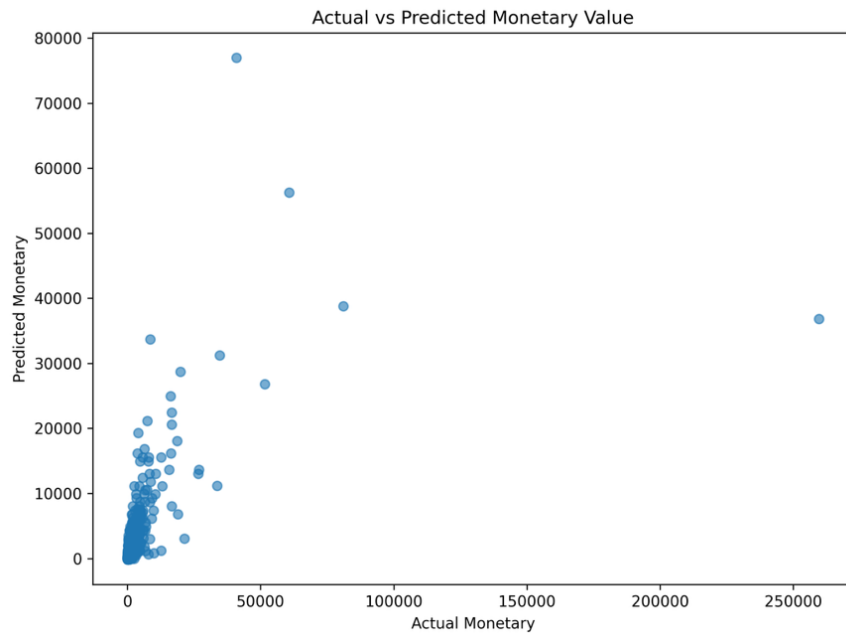
Model	Features	Target	Freq Limit	Mont Limit	MAE	RMSE	MAPE	R2
Linear Regression	Recency, Frequency	Monetary			1451.81	8199.76	99.64	0.3437
Filtered Linear Regression	Recency, Frequency	Monetary	50	25000	666.49	1460.24	94.89	0.5412
Extended Feature Regression	Recency, Frequency, ProductDiversity, AvgDaysBetweenTransactions	Monetary			1485.11	8323.77	116.12	0.3237

Regression Performance & Outlier Effects

Removing extreme spending outliers improved prediction accuracy.

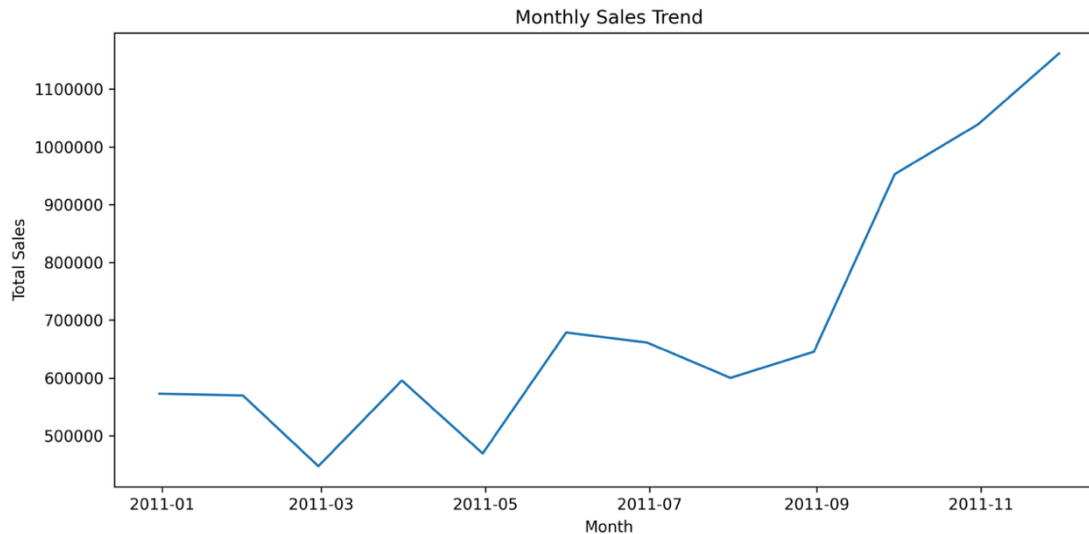
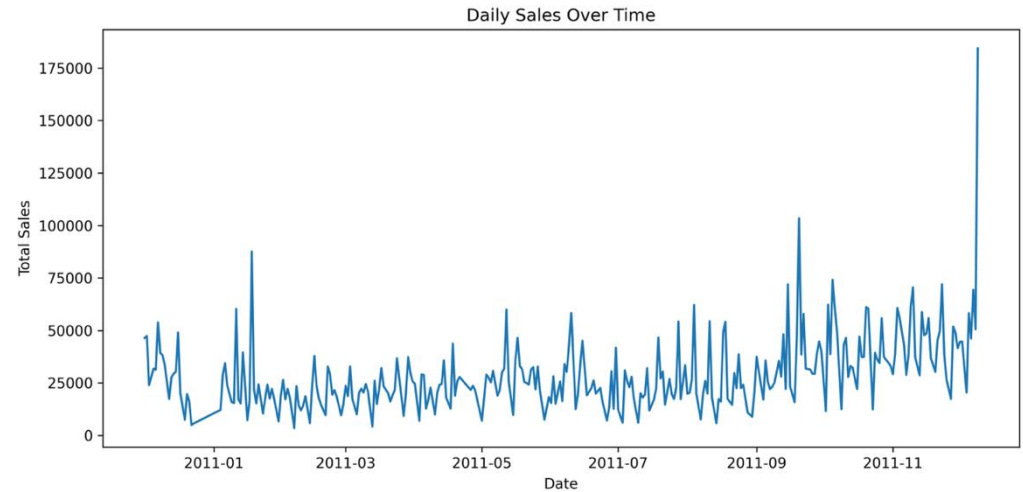
R^2 Values Comparison

- R^2 was 0.34 before filtering
- R^2 increased to 0.54 after filtering



Time Series & Forecasting

- Daily sales showed high volatility
- Monthly aggregation revealed upward trend
- ARIMA(4,1,1) used for forecasting



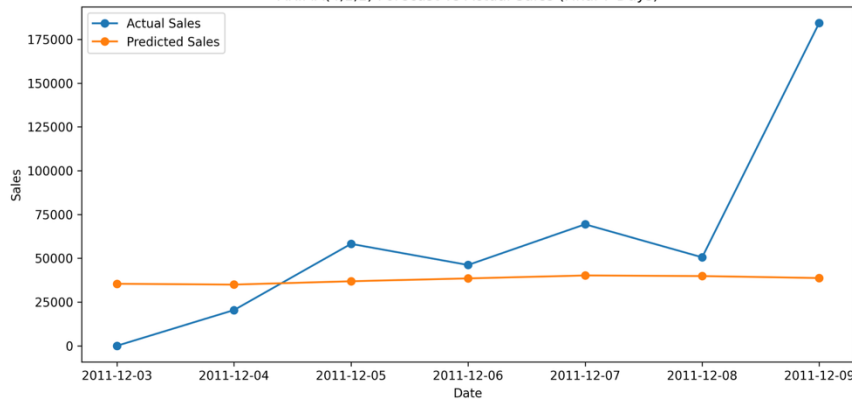
- Extreme spikes increased error
- Forecasts smoothed short-term fluctuations
- Stronger performance during stable periods

Time Series & Forecasting

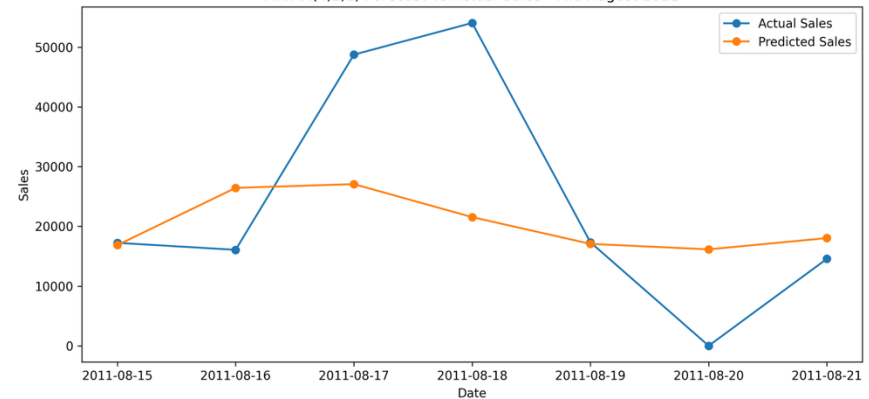
- Performance improved during stable periods
- Extreme spikes caused large forecast errors
- ARIMA captured trends better than sudden demand changes

Model	Validation Period	MAE	RMSE	MAPE
ARIMA(4,1,1)	2011-12-03 to 2011-12-09	37,795.95	58,758.22	44.52
ARIMA(4,1,1)	2011-08-15 to 2011-08-21	12,122.77	16,515.02	32.78

ARIMA(4,1,1) Forecast vs Actual Sales (Final 7 Days)



ARIMA(4,1,1) Forecast vs Actual Sales - Mid-August 2011



Association Rules

- Apriori identified products frequently purchased together
- Strongest rules appeared within customer Cluster 2
- Tea set products showed extremely high co-purchase behavior
- Results support recommendation systems and product bundling

antecedents	consequents	support	confidence	lift	Cluster
GREEN REGENCY TEACUP AND SAUCER ROSES REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER	0.0221	0.7299	22.7867	0
PINK REGENCY TEACUP AND SAUCER ROSES REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.0221	0.8664	22.6436	0
GREEN REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER	0.0241	0.6224	22.506	1
GREEN REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	0.0273	0.7041	18.7582	1
PINK REGENCY TEACUP AND SAUCER ROSES REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.0202	0.9355	26.8016	2
GREEN REGENCY TEACUP AND SAUCER PINK REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	0.0202	0.8529	21.5302	2

The strongest association rule appeared in Cluster 2.

- The support value of 0.0202 indicates that approximately 2.02% of all transactions contained these products together.
- The high confidence value of 93.6% indicates that customers who purchased the Pink Regency Teacup and Saucer and Roses Regency Teacup and Saucer were very likely to also purchase the Green Regency Teacup and Saucer.
- A lift value of 26.8 means customers were about 26.8 times more likely to purchase these products together compared to random chance.

Novelty / Non-Triviality / Usefulness

- **Novelty**
 - Combined clustering, forecasting, regression, and association rule mining
 - Compared traditional RFM features against extended behavioral variables
 - Used quantitative evaluation metrics for validation
- **Challenges**
 - Highly skewed spending behavior
 - Cluster selection and interpretation
 - Forecasting extreme sales spikes
 - Extended features did not improve performance
- **Risks & Limitations**
 - Misclassification may lead to ineffective customer targeting
 - Forecasting errors may impact inventory planning
 - Dataset primarily represents UK retail customers
 - High-value customers remained difficult to model accurately
- **Business Value**
 - Customer segmentation for targeted marketing
 - Product recommendations and bundling
 - Inventory planning and sales monitoring
 - Support for data-driven retail decision-making

Ethical Considerations & Limitations

- **Data Transparency**
 - UCI Online Retail dataset
 - Publicly available research dataset
 - No web scraping or third-party API collection
 - Used in compliance with educational/research purposes
- **Privacy & PII**
 - No direct PII (names, addresses, payment data)
 - CustomerID functioned as a quasi-identifier
 - Analysis performed only at aggregated customer level
- **Bias & Representativeness**
 - Dataset primarily represents UK retail customers
 - Customer spending highly skewed toward a small number of high-value customers
 - Log transformation and scaling used to reduce skew effects
- **Potential Consequences**
 - Misclassification could lead to ineffective customer targeting
 - Forecasting errors could affect inventory planning and promotions
 - High-value customers remained difficult to model accurately

Conclusion & Future Work

- **Key Findings**

- RFM features produced the strongest clustering performance
- $k=2$ achieved the best cluster separation
- Regression performance improved after removing extreme outliers
- ARIMA captured trends but struggled with sales spikes
- Association rules identified strong product co-purchasing behavior

- **Future Work**

- Seasonal and advanced forecasting models
- Deep learning approaches
- Real-time recommendation systems
- Additional behavioral feature exploration
- Investigate whether outlier filtering improves clustering and forecasting performance, similar to the regression results
 - Outlier filtering improved regression but slightly weakened clustering separation.